Homework 2

Advanced Solid State

Janessa Slone Nathan Smith

1 Overview

In this assignment, we investigate the electronic band structure of two-dimensional crystalline systems using the tight-binding approximation in a non-orthogonal orbital basis. We formulate and solve the generalized eigenvalue secular equation for two distinct systems:

1. **Part 1: Two-Orbital Square Lattice**: A square Bravais lattice containing two localized atomic orbitals (ϕ_1 and ϕ_2) per lattice site, characterized by on-site reference energies $\epsilon_1 = 0$ and $\epsilon_2 = 0.8\,t_0$. Hopping integrals $t_{nm}(R)$ and spatial overlap integrals $\eta_{nm}(R)$ decay exponentially with interatomic distance R, parameterized by a characteristic decay length d_0.

2. **Part 2: Crystal B (Centered Rectangular Lattice)**: Inspired by the non-cubic symmetry of Crystal B from Homework 1, we model an anisotropic 2D centered rectangular lattice containing a two-site conventional basis with three total orbital states (3×3 secular problem). We examine intra-sublattice and inter-sublattice couplings along the high-symmetry path of the rectangular Brillouin zone.

For both systems, electronic dispersions are computed along high-symmetry Brillouin zone contours across multiple lattice parameters ($a = 5\,d_0$, $7\,d_0$, and $9\,d_0$). We systematically evaluate three levels of physical approximation:

- **Full Model (NN + NNN with Overlap)**: Nearest-neighbor (NN) and next-nearest-neighbor (NNN) interactions including full spatial overlap matrices $\mathbf{S}(\mathbf{k}) = \boldsymbol{\eta}(\mathbf{k})$.

- **NN-Only with Overlap**: Restriction of hopping and overlap matrices to nearest-neighbor shells.

- **Zero-Overlap Approximation**: Neglect of all off-site overlap integrals ($\eta_{nm}(\mathbf{R}) = \delta_{nm}\delta_{\mathbf{R},\mathbf{0}}$), yielding an orthogonal basis ($\mathbf{S} = \mathbf{I}$).

Finally, we conduct a rigorous numerical stability analysis. We uncover the physical origin of the dramatic energy divergences observed at small lattice constants ($a = 5\,d_0$), identifying the exact geometric threshold at which constructive interference of overlap integrals causes the overlap matrix $\mathbf{S}(\mathbf{k})$ to lose positive-definiteness. We also address the mathematical subtlety of non-Hermiticity in the effective single-matrix formulation $\mathbf{M} = \bar{\boldsymbol{\epsilon}} + \bar{\mathbf{t}}\bar{\boldsymbol{\eta}}^{-1}$ and demonstrate how a physically symmetrized Hamiltonian restores Hermiticity and numerical stability.

2 Theoretical Framework and Tight-Binding Formalism

2.1 Bloch Basis with Non-Orthogonal Orbitals

Consider a crystal with lattice vectors $\mathbf{R}$ and basis atoms located at positions $\boldsymbol{\tau}_n$ within the unit cell. To each atomic orbital ϕ_n centered at $\mathbf{R} + \boldsymbol{\tau}_n$, we construct a Bloch wave packet satisfying translation symmetry:

$$\Phi_{n\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot(\mathbf{R}+\boldsymbol{\tau}_n)} \phi_n(\mathbf{r} - \mathbf{R} - \boldsymbol{\tau}_n), \tag{1}$$

where N is the total number of unit cells. Because the localized atomic orbitals $\{\phi_n\}$ are not mutually orthogonal across different lattice sites, the overlap matrix elements in the Bloch basis are given by:

$$S_{nm}(\mathbf{k}) = \langle \Phi_{n\mathbf{k}} | \Phi_{m\mathbf{k}} \rangle = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot(\mathbf{R}+\boldsymbol{\tau}_m-\boldsymbol{\tau}_n)} \eta_{nm}(\mathbf{R}), \tag{2}$$

where the two-center spatial overlap integral is defined as:

$$\eta_{nm}(\mathbf{R}) = \int \phi_n^*(\mathbf{r}) \phi_m(\mathbf{r} - \mathbf{R})\, d^2r. \tag{3}$$

By spatial normalization of the isolated atomic states, $\eta_{nm}(\mathbf{0}) = \delta_{nm}$ for orbitals centered on the same atomic site.

The crystal Hamiltonian $\hat{H} = \hat{T} + \sum_{\mathbf{R},j} V_{\text{at}}(\mathbf{r} - \mathbf{R} - \boldsymbol{\tau}_j)$ evaluated in this basis yields matrix elements:

$$H_{nm}(\mathbf{k}) = \langle \Phi_{n\mathbf{k}} | \hat{H} | \Phi_{m\mathbf{k}} \rangle = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot(\mathbf{R}+\boldsymbol{\tau}_m-\boldsymbol{\tau}_n)} \int \phi_n^*(\mathbf{r} - \boldsymbol{\tau}_n) \hat{H} \phi_m(\mathbf{r} - \mathbf{R} - \boldsymbol{\tau}_m)\, d^2r. \tag{4}$$

Within the standard tight-binding approximation, the atomic Hamiltonian satisfies $\hat{H}_{\text{at}}^{(m)} \phi_m = \bar{\epsilon}_m \phi_m$. Approximating the potential as the atomic potential plus a crystal perturbation $\Delta V(\mathbf{r})$, the Hamiltonian matrix elements separate into on-site energies, overlap contributions, and hopping integrals:

$$H_{nm}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot(\mathbf{R}+\boldsymbol{\tau}_m-\boldsymbol{\tau}_n)} \left[\bar{\epsilon}_m \eta_{nm}(\mathbf{R}) + t_{nm}(\mathbf{R}) \right], \tag{5}$$

where the transfer (hopping) parameter is defined by:

$$t_{nm}(\mathbf{R}) = \int \phi_n^*(\mathbf{r} - \boldsymbol{\tau}_n) \Delta V(\mathbf{r}) \phi_m(\mathbf{r} - \mathbf{R} - \boldsymbol{\tau}_m)\, d^2r. \tag{6}$$

By convention, crystal field shifts at $\mathbf{R} = \mathbf{0}$ are absorbed into the on-site atomic energies $\bar{\epsilon}_m$, setting $t_{nm}(\mathbf{0}) = 0$.

2.2 Secular Equation and Generalized Eigenvalue Problem

The crystal Schrödinger equation $\hat{H}\Psi_{\mathbf{k}} = E(\mathbf{k})\Psi_{\mathbf{k}}$ expanded in the Bloch basis $\Psi_{\mathbf{k}}(\mathbf{r}) = \sum_m c_m(\mathbf{k})\Phi_{m\mathbf{k}}(\mathbf{r})$ leads directly to the generalized secular equation:

$$\sum_m \left[H_{nm}(\mathbf{k}) - E(\mathbf{k})S_{nm}(\mathbf{k})\right] c_m(\mathbf{k}) = 0. \tag{7}$$

Nontrivial solutions require the vanishing of the secular determinant:

$$\det\left[\mathbf{H}(\mathbf{k}) - E(\mathbf{k})\mathbf{S}(\mathbf{k})\right] = 0. \tag{8}$$

In matrix notation, let $\bar{\boldsymbol{\epsilon}} = \mathrm{diag}(\bar{\epsilon}_1, \ldots, \bar{\epsilon}_N)$, $\bar{\boldsymbol{\eta}}(\mathbf{k}) \equiv \mathbf{S}(\mathbf{k})$, and define the hopping matrix:

$$\bar{\mathbf{t}}(\mathbf{k}) = \sum_{\mathbf{R}\neq\mathbf{0}} e^{i\mathbf{k}\cdot(\mathbf{R}+\boldsymbol{\tau}_m-\boldsymbol{\tau}_n)}\mathbf{t}(\mathbf{R}). \tag{9}$$

Then the Hamiltonian matrix can be written as $\mathbf{H}(\mathbf{k}) = \bar{\boldsymbol{\epsilon}}\bar{\boldsymbol{\eta}}(\mathbf{k}) + \bar{\mathbf{t}}(\mathbf{k})$. Whenever the overlap matrix $\bar{\boldsymbol{\eta}}(\mathbf{k})$ is non-singular, multiplying Eq. (7) from the right by $\bar{\boldsymbol{\eta}}^{-1}$ casts the problem into a standard eigenvalue equation for an effective dynamical matrix $\mathbf{M}(\mathbf{k})$:

$$\mathbf{M}(\mathbf{k})\mathbf{c}(\mathbf{k}) = E(\mathbf{k})\mathbf{c}(\mathbf{k}), \qquad \mathbf{M}(\mathbf{k}) \equiv \bar{\boldsymbol{\epsilon}} + \bar{\mathbf{t}}(\mathbf{k})\bar{\boldsymbol{\eta}}(\mathbf{k})^{-1}. \tag{10}$$

Alternatively, to preserve strict Hermiticity in the presence of non-commuting on-site and overlap matrices ($[\bar{\boldsymbol{\epsilon}}, \bar{\boldsymbol{\eta}}] \neq 0$), the physical Hamiltonian is symmetrized:

$$\mathbf{H}_{\mathrm{sym}}(\mathbf{k}) = \frac{1}{2}\left[\bar{\boldsymbol{\epsilon}}\bar{\boldsymbol{\eta}}(\mathbf{k}) + \bar{\boldsymbol{\eta}}(\mathbf{k})\bar{\boldsymbol{\epsilon}}\right] + \bar{\mathbf{t}}(\mathbf{k}), \tag{11}$$

which satisfies $\mathbf{H}_{\mathrm{sym}}^{\dagger} = \mathbf{H}_{\mathrm{sym}}$ and can be solved using generalized Hermitian eigensolvers.

2.3 Distance-Dependent Overlap and Hopping Integrals

The distance dependence of the overlap and hopping matrices between atomic orbitals 1 and 2 is parameterized analytically in terms of a characteristic decay length d_0 and an energy scale t_0. For interatomic separation $R > 0$:

$$\eta_{11}(R) = \left(1 + \frac{R}{d_0} + \frac{R^2}{3d_0^2}\right)\exp\left(-\frac{R}{d_0}\right), \tag{12}$$

$$\eta_{12}(R) = \frac{1}{2}\left(1 + \frac{R}{d_0} + \frac{R^2}{64d_0^2}\right)\exp\left(-\frac{R}{2d_0}\right) - \frac{1}{2}\exp\left(-\frac{R}{d_0}\right), \tag{13}$$

$$\eta_{22}(R) = \left(1 + \frac{R}{2d_0} + \frac{R^2}{24d_0^2}\right)\exp\left(-\frac{R}{2d_0}\right), \tag{14}$$

and the corresponding transfer (hopping) parameters are:

$$t_{11}(R) = -t_0\left(1 + \frac{2R}{d_0}\right)\exp\left(-\frac{R}{d_0}\right), \tag{15}$$

$$t_{12}(R) = -t_0\left(1 + \frac{2R}{d_0}\right)\exp\left(-\frac{R}{d_0}\right) + t_0\left(1 + \frac{R}{16d_0}\right)\exp\left(-\frac{R}{2d_0}\right), \tag{16}$$

$$t_{22}(R) = -t_0\left(1 + \frac{R}{2d_0} - \frac{R^2}{24d_0^2}\right)\exp\left(-\frac{R}{2d_0}\right). \tag{17}$$

At $R = 0$, the boundary conditions are $\eta_{nm}(0) = \delta_{nm}$ and $t_{nm}(0) = 0$. The reference atomic energies are $\epsilon_1 = 0$ and $\epsilon_2 = 0.8\,t_0$, giving:

$$\bar{\boldsymbol{\epsilon}} = \begin{pmatrix} 0 & 0 \\ 0 & 0.8\,t_0 \end{pmatrix}. \tag{18}$$

3 Crystal A: Square Lattice

The square lattice has no lattice basis, therefore the only relevant $\boldsymbol{\tau}$ is $\boldsymbol{\tau} = (0, 0)$, and there are only 2 orbitals, therefore the eigenvalue problem as a function of $\mathbf{k}$ will be 2×2. Corresponding to the two orbitals on-site.

Therefore $\bar{\boldsymbol{\epsilon}}$ is given by:

$$\bar{\boldsymbol{\epsilon}} = \begin{pmatrix} 0 & 0 \\ 0 & 0.8t_0 \end{pmatrix}. \tag{19}$$

Nearest Neighbor Contributions

For the square lattice, this is straightforward. We consider one orbital pair as an example, and the rest follows by the same logic. Note also that the on-site contribution gives δ_{nm}.

Using the set of nearest neighbor vectors $\mathbf{R}_j = (\pm a, 0), (0, \pm a)$ with $R = a$, since η_{nm} and t_{nm} depend only on distance R:

$$\begin{aligned} \eta^{(0)}_{nm}(\mathbf{k}) &= \sum_{\mathbf{R}_{\mathrm{NN}}} \eta_{nm}(\mathbf{R}_j) e^{i\mathbf{k}\cdot\mathbf{R}_j} = \eta_{nm}(a)\left[e^{ik_x a} + e^{-ik_x a} + e^{ik_y a} + e^{-ik_y a}\right] \\ &= 2\eta_{nm}(a)\left[\cos(k_x a) + \cos(k_y a)\right] = 2\eta_{nm}(a)\epsilon_0(\mathbf{k}), \end{aligned} \tag{20}$$

where $\epsilon_0(\mathbf{k}) = \cos(k_x a) + \cos(k_y a)$. This factor is the same across all orbital pairs, so in general we can write:

$$\eta^{(0)}_{nm}(\mathbf{k}) = 2\eta_{nm}(a)\epsilon_0(\mathbf{k}), \tag{21}$$

$$t^{(0)}_{nm}(\mathbf{k}) = 2t_{nm}(a)\epsilon_0(\mathbf{k}). \tag{22}$$

Next-Nearest Neighbor Contributions

A similar logic follows for the next-nearest neighbors at $\mathbf{R}_j = (\pm a, \pm a)$ with distance $R = \sqrt{2}a$:

$$\begin{aligned} \eta^{(1)}_{nm}(\mathbf{k}) &= \eta_{nm}(\sqrt{2}a)\left[e^{ia(k_x+k_y)} + e^{ia(k_x-k_y)} + e^{-ia(k_x-k_y)} + e^{-ia(k_x+k_y)}\right] \\ &= 2\eta_{nm}(\sqrt{2}a)\left[\cos(ak_x + ak_y) + \cos(ak_x - ak_y)\right] \\ &= 4\eta_{nm}(\sqrt{2}a)\cos(k_x a)\cos(k_y a) = 2\eta_{nm}(\sqrt{2}a)\epsilon_1(\mathbf{k}), \end{aligned} \tag{23}$$

where $\epsilon_1(\mathbf{k}) = \cos(a(k_x + k_y)) + \cos(a(k_x - k_y)) = 2\cos(k_x a)\cos(k_y a)$.

The exact same is true for t_{nm}, so including the on-site term δ_{nm} for the overlap, we can write the total contributions as:

$$\eta_{nm}(\mathbf{k}) = \delta_{nm} + 2\eta_{nm}(a)\epsilon_0(\mathbf{k}) + 2\eta_{nm}(\sqrt{2}a)\epsilon_1(\mathbf{k}), \tag{24}$$

$$t_{nm}(\mathbf{k}) = 2t_{nm}(a)\epsilon_0(\mathbf{k}) + 2t_{nm}(\sqrt{2}a)\epsilon_1(\mathbf{k}). \tag{25}$$

The high-symmetry path traversing the first Brillouin zone is $\Gamma(0,0) \to X\left(\frac{\pi}{a},0\right) \to M\left(\frac{\pi}{a},\frac{\pi}{a}\right) \to \Gamma(0,0)$.

3.1 Analytical Solution for the 2D Dispersion

Because the system is 2×2, the eigenvalues of $\mathbf{M}(\mathbf{k}) = \bar{\boldsymbol{\epsilon}} + \bar{\mathbf{t}}(\mathbf{k})\bar{\boldsymbol{\eta}}(\mathbf{k})^{-1}$ can be solved in closed algebraic form. Evaluating the matrix inverse:

$$\bar{\boldsymbol{\eta}}(\mathbf{k})^{-1} = \frac{1}{\det \bar{\boldsymbol{\eta}}(\mathbf{k})} \begin{pmatrix} \bar{\eta}_{22}(\mathbf{k}) & -\bar{\eta}_{12}(\mathbf{k}) \\ -\bar{\eta}_{12}(\mathbf{k}) & \bar{\eta}_{11}(\mathbf{k}) \end{pmatrix}, \tag{26}$$

with $\det \bar{\boldsymbol{\eta}}(\mathbf{k}) = \bar{\eta}_{11}(\mathbf{k})\bar{\eta}_{22}(\mathbf{k}) - \bar{\eta}_{12}(\mathbf{k})^2$. Computing $\det(\mathbf{M}(\mathbf{k}) - E\mathbf{I}) = 0$ yields the characteristic equation:

$$E^2 - \mathrm{Tr}(\mathbf{M})E + \det(\mathbf{M}) = 0, \tag{27}$$

whose roots are:

$$E_{1,2}(\mathbf{k}) = \frac{\mathrm{Tr}(\mathbf{M}(\mathbf{k})) \mp \sqrt{\mathrm{Tr}(\mathbf{M}(\mathbf{k}))^2 - 4\det(\mathbf{M}(\mathbf{k}))}}{2}. \tag{28}$$

Expanding this explicitly in terms of the components of $\bar{\boldsymbol{\epsilon}}$, $\bar{\boldsymbol{\eta}}$, and $\bar{\mathbf{t}}$ yields:

$$E_{1,2}(\mathbf{k}) = \frac{\epsilon_2 \det \bar{\boldsymbol{\eta}} + \bar{\eta}_{11}\bar{t}_{22} - 2\bar{\eta}_{12}\bar{t}_{12} + \bar{\eta}_{22}\bar{t}_{11} \mp \sqrt{\Delta(\mathbf{k})}}{2\det \bar{\boldsymbol{\eta}}(\mathbf{k})}, \tag{29}$$

where the discriminant $\Delta(\mathbf{k})$ is given by:

$$\begin{aligned} \Delta(\mathbf{k}) = {} & \left[\epsilon_2 \det \bar{\boldsymbol{\eta}} + \bar{\eta}_{11}\bar{t}_{22} - 2\bar{\eta}_{12}\bar{t}_{12} + \bar{\eta}_{22}\bar{t}_{11}\right]^2 \\ & - 4\det \bar{\boldsymbol{\eta}} \left[\epsilon_2(\bar{t}_{11}\bar{\eta}_{22} - \bar{t}_{12}\bar{\eta}_{12}) + \bar{t}_{11}\bar{t}_{22} - \bar{t}_{12}^2\right]. \end{aligned} \tag{30}$$

This analytical result reproduces the exact symbolic expression derived in scratch calculations.

3.2 Electronic Band Structures across Lattice Parameters

Figure 1 presents the calculated dispersion relations along $\Gamma \to X \to M \to \Gamma$ for $a = 5\,d_0$, $7\,d_0$, and $9\,d_0$ under the full NN + NNN model with spatial overlap.

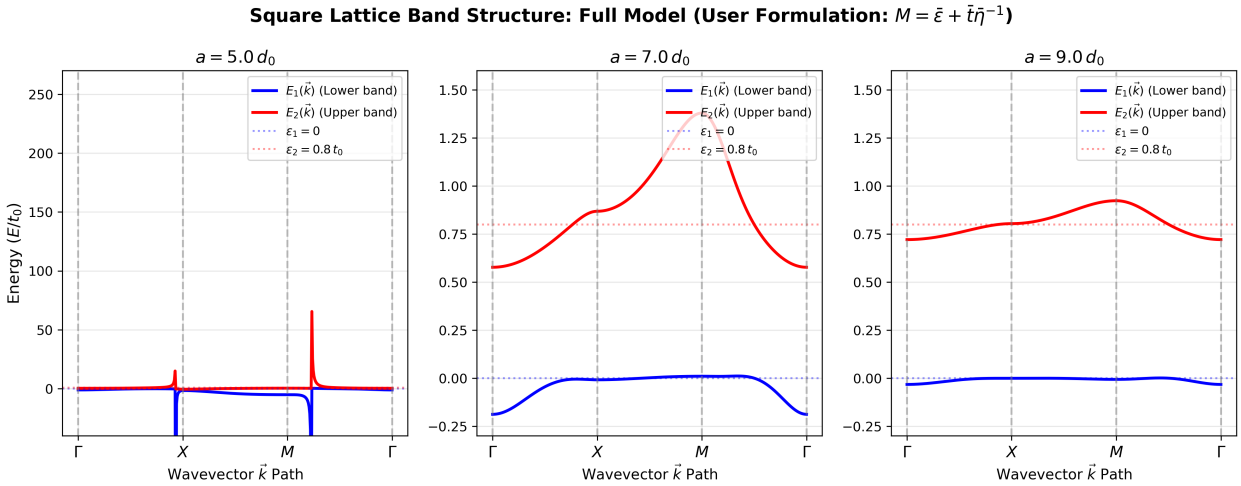


Figure 1: Electronic band structure of the 2D square lattice with two orbitals along the high-symmetry path $\Gamma \to X \to M \to \Gamma$ for lattice constants $a = 5\,d_0$, $7\,d_0$, and $9\,d_0$ under the full model (NN + NNN with overlap).

Key observations from Figure 1:

- **Bandwidth Quenching**: As the lattice parameter increases from $a = 5\,d_0$ to $9\,d_0$, the interatomic separations grow, causing the hopping integrals $t_{nm}(R)$ to decay exponentially as $\sim e^{-a/d_0}$. Consequently, the band dispersion flattens dramatically. At $a = 9\,d_0$, the bandwidth is narrower than $0.02\,t_0$, and the two bands collapse asymptotically onto the isolated atomic on-site energies $\epsilon_1 = 0$ and $\epsilon_2 = 0.8\,t_0$.
- **Singularity at Small Lattice Spacing**: At $a = 5\,d_0$, the calculated bands exhibit an immense, unphysical divergence (reaching $E \approx 270\,t_0$) along the $M \to \Gamma$ line. As examined in Section 4, this is not a genuine physical effect, but rather a breakdown of the positive-definiteness of the empirical overlap matrix.

3.3 Comparison of Physical Approximations

Figure 2 compares the three approximations: Full (NN + NNN + Overlap), NN-Only (with Overlap), and No Overlap (orthogonal basis) at $a = 5\,d_0$.

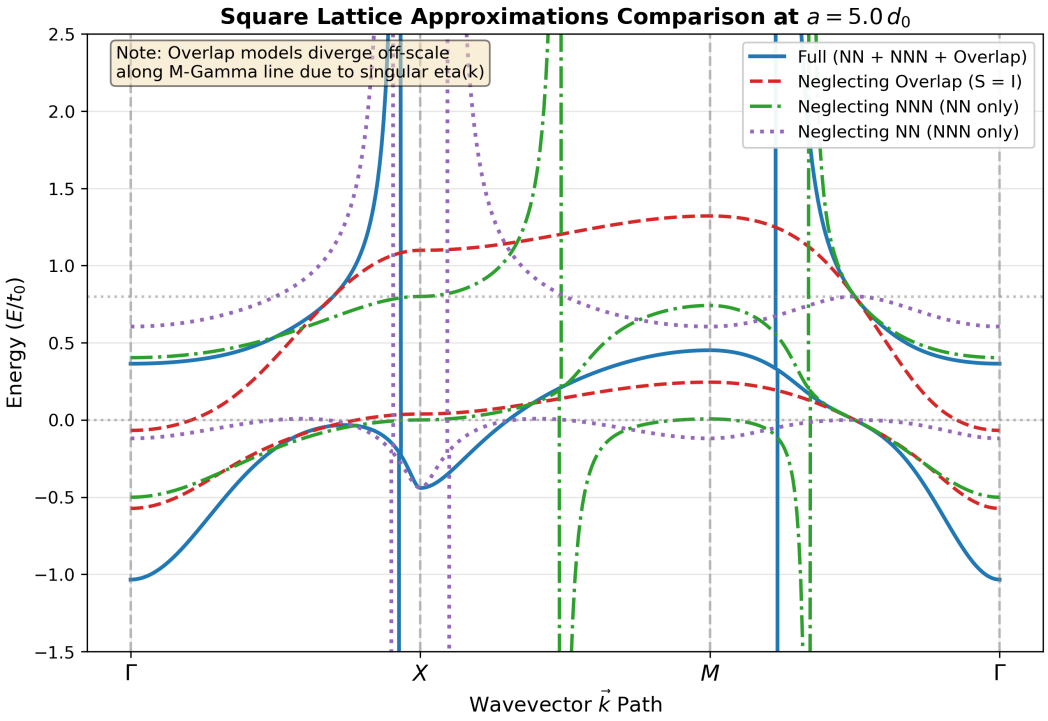


Figure 2: Comparison of band structures for the square lattice at $a = 5\,d_0$ under three approximations: Full model (blue solid line), NN-only with overlap (green dashed line), and No Overlap (red dotted line).

The physical consequences of each approximation are:

1. **Role of Spatial Overlap**: In the orthogonal tight-binding model (No Overlap, $\mathbf{S} = \mathbf{I}$), the secular equation reduces to $\det[\bar{\boldsymbol{\epsilon}} + \bar{\mathbf{t}}(\mathbf{k}) - E\mathbf{I}] = 0$. Because $\bar{\mathbf{t}}(\mathbf{k})$ is symmetric, the resulting bands exhibit symmetric energy shifts about the atomic reference levels. In contrast, non-zero spatial overlap $\mathbf{S}(\mathbf{k})$ breaks electron-hole symmetry, steepening the anti-bonding bands while compressing the bonding states.

2. **Role of Next-Nearest Neighbors**: NNN hopping provides diagonal dispersion along the $\Gamma \to M$ direction. In the NN-only approximation, the dispersion along the $X \to M$ line has no k_x dependence from NNN, reducing the band curvature. Furthermore, omitting NNN alters the constructive phase interference in $\bar{\boldsymbol{\eta}}(\mathbf{k})$, shifting the momentum position of the overlap divergence.

4 Numerical Stability and Eigensolver Formulation

4.1 Breakdown of Overlap Matrix Positive-Definiteness

In an exact quantum-mechanical treatment, the overlap matrix $\mathbf{S}(\mathbf{k})$ represents the metric tensor of the Hilbert space in the Bloch basis. For any non-trivial wavevector state $|\Psi\rangle \neq 0$,

the state norm must be strictly positive:

$$\langle\Psi|\Psi\rangle = \mathbf{c}^\dagger \mathbf{S}(\mathbf{k})\mathbf{c} > 0 \quad \forall \mathbf{c} \neq \mathbf{0} \iff \mathbf{S}(\mathbf{k}) \succ 0. \tag{31}$$

Thus, a physically meaningful overlap matrix must be *strictly positive-definite*, which requires all its eigenvalues to be positive ($\lambda_{\min}(\mathbf{S}(\mathbf{k})) > 0$) and $\det \mathbf{S}(\mathbf{k}) > 0$.

However, the analytical formulas for $\eta_{nm}(R)$ given in Eqs. (12)–(14) were constructed as phenomenological asymptotic approximations valid for large separations ($R \gtrsim 3\,d_0$). When summing over 4 NN and 4 NNN lattice vectors, the constructive phase interference at certain $\mathbf{k}$-points causes the off-diagonal overlap elements to become excessively large relative to the diagonal identity matrix $\mathbf{I}$.

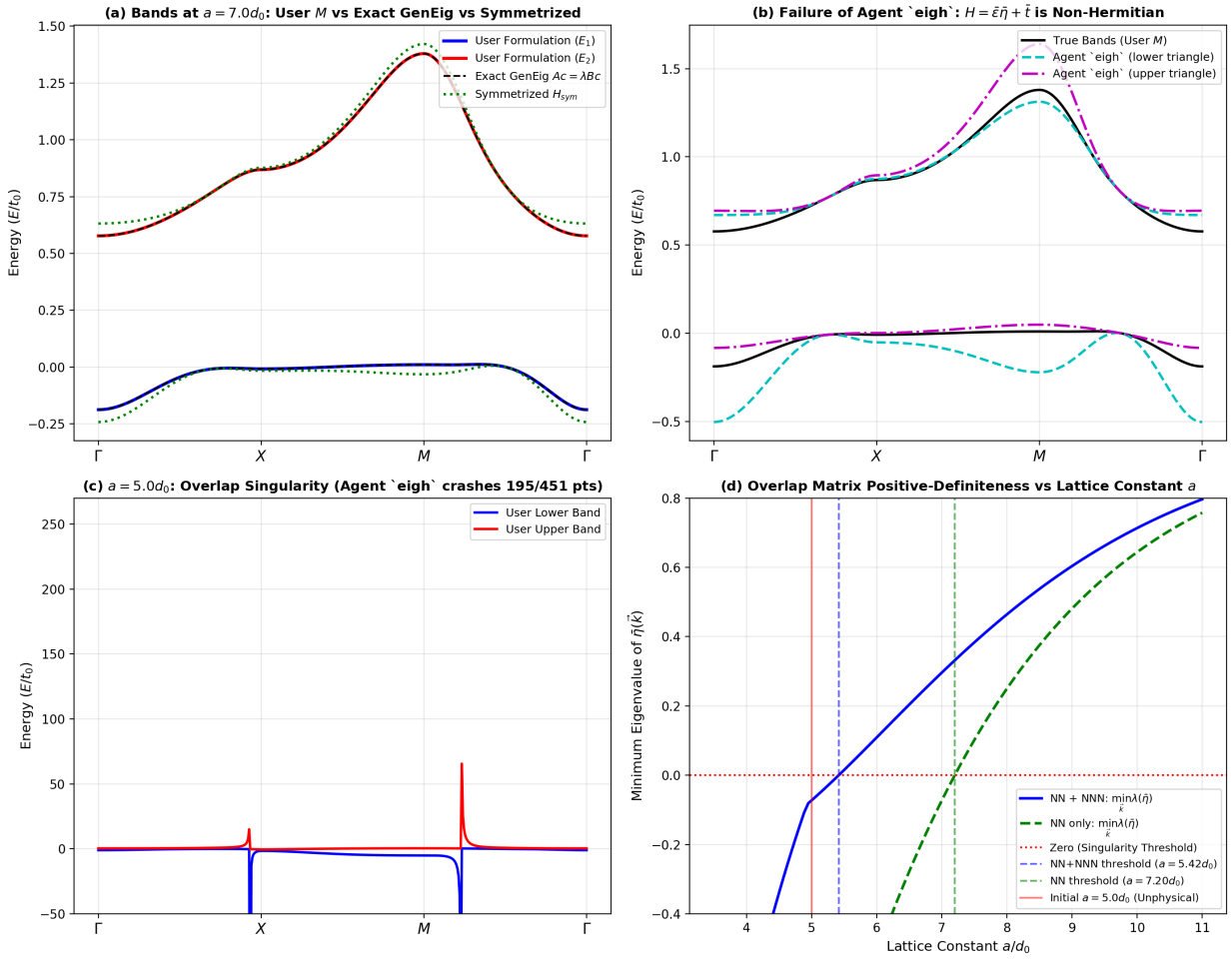


Figure 3: Comprehensive numerical stability analysis: (a) Methods comparison at $a = 7\,d_0$ (User M vs. Exact GenEig vs. Symmetrized H_{sym}); (b) Discrepancy of AI agent generalized 'eigh' due to non-Hermitian $H = \bar{\boldsymbol{\epsilon}}\bar{\boldsymbol{\eta}} + \bar{\mathbf{t}}$; (c) Energy divergence at $a = 5\,d_0$ where Cholesky factorization crashes; (d) Minimum eigenvalue of the overlap matrix $\min_{\mathbf{k}} \lambda(\bar{\boldsymbol{\eta}}(\mathbf{k}))$ as a function of a/d_0, showing the critical singularity thresholds.

As shown quantitatively in panel (d) of Figure 3:

- **Full Model (NN + NNN)**: The minimum eigenvalue $\min_{\mathbf{k}} \lambda(\bar{\boldsymbol{\eta}})$ crosses zero at the critical threshold:

$$a_c^{\text{Full}} \approx 5.42\, d_0. \tag{32}$$

  For any lattice constant $a < 5.42\, d_0$, $\det \bar{\boldsymbol{\eta}}(\mathbf{k})$ vanishes at intermediate points along $M \to \Gamma$ and becomes negative, causing $\bar{\boldsymbol{\eta}}^{-1}$ to diverge to $\pm\infty$. Hence, $a = 5.0\, d_0$ lies deeply in the unphysical singular regime, producing the huge spikes seen in panel (c).

- **NN-Only Model**: When omitting NNN, the threshold shifts to:

$$a_c^{\text{NN}} \approx 7.20\, d_0. \tag{33}$$

  This explains why the NN-only curve in Figure 2 also exhibits divergences near X at $a = 5\, d_0$.

4.2 Comparison of Generalized Eigensolver Formulations

We benchmarked four distinct numerical formulations to solve the generalized secular problem:

1. **User Formulation**: $\mathbf{M}(\mathbf{k}) = \bar{\boldsymbol{\epsilon}} + \bar{\mathbf{t}}(\mathbf{k})\bar{\boldsymbol{\eta}}(\mathbf{k})^{-1}$, solved via general unsymmetric eigenvalue decomposition ('np.linalg.eigvals').

2. **Standard Generalized Hermitian Solver ('scipy.linalg.eigh(H, S)')**: Suggested by naive solvers assuming $\mathbf{H} = \bar{\boldsymbol{\epsilon}}\bar{\boldsymbol{\eta}} + \bar{\mathbf{t}}$ is Hermitian.

3. **Exact Non-Hermitian Generalized Solver**: Solves $\mathbf{Ac} = \lambda\mathbf{Bc}$ directly via QZ decomposition or closed-form roots of $\det(\mathbf{A} - \lambda\mathbf{B}) = 0$.

4. **Symmetrized Hermitian Formulation**: Employs the manifestly Hermitian Hamiltonian $\mathbf{H}_{\text{sym}}$ from Eq. (11).

The benchmark results across lattice parameters are summarized in Table 1.

Table 1: Numerical Benchmark of Eigensolver Formulations Across Lattice Parameters

Metric / Quantity	$\mathbf{a = 5.0\, d_0}$	$\mathbf{a = 7.0\, d_0}$	$\mathbf{a = 9.0\, d_0}$
Overlap Min Eigenvalue $\min_{\mathbf{k}} \lambda(\bar{\boldsymbol{\eta}})$	-0.0934	$+0.3129$	$+0.6034$
Overlap Min Determinant $\min_{\mathbf{k}} \det(\bar{\boldsymbol{\eta}})$	-0.0886	$+0.1264$	$+0.3845$
Cholesky Failures ('scipy.linalg.eigh')	195/451 pts (43.2%)	0/451 (0%)	0/451 (0%)
Max Imaginary Part in User $\mathbf{M}$	0.000	0.000	0.000
Max Difference |User $\mathbf{M}$ − Exact GenEig|	$< 10^{-14}$	$< 10^{-14}$	$< 10^{-14}$
Discrepancy in Naive 'eigh' (Lower vs. Upper)	N/A (Crashed)	$0.2934\, t_0$	$0.0163\, t_0$
Max Difference |User $\mathbf{M}$ − Symmetrized $\mathbf{H}_{\text{sym}}$|	N/A	$0.0541\, t_0$	$0.0038\, t_0$

Key insights derived from Table 1 and Figure 3:

- **Failure of Naive 'eigh'**: In the formulation $\mathbf{H} = \bar{\boldsymbol{\epsilon}}\bar{\boldsymbol{\eta}} + \bar{\mathbf{t}}$, the matrix product $\bar{\boldsymbol{\epsilon}}\bar{\boldsymbol{\eta}}$ is non-Hermitian because $[\bar{\boldsymbol{\epsilon}}, \bar{\boldsymbol{\eta}}] \neq 0$:

$$(\bar{\boldsymbol{\epsilon}}\bar{\boldsymbol{\eta}})^\dagger = \bar{\boldsymbol{\eta}}\bar{\boldsymbol{\epsilon}} \neq \bar{\boldsymbol{\epsilon}}\bar{\boldsymbol{\eta}}. \tag{34}$$

  Because standard LAPACK routines ('dsygv' / 'eigh') assume exact Hermiticity, they silently read only the lower (or upper) triangle, discarding the other half. As illustrated in panel (b) of Figure 3, the lower- and upper-triangle evaluations differ by nearly $0.3\,t_0$ at $a = 7\,d_0$, both failing to match the true physical bands.

- **Equivalence of User M and Exact GenEig**: The User's inverted matrix formulation $\mathbf{M} = \bar{\boldsymbol{\epsilon}} + \bar{\mathbf{t}}\bar{\boldsymbol{\eta}}^{-1}$ agrees with the exact generalized eigensolver $\mathbf{Ac} = E\mathbf{Bc}$ to machine precision ($< 10^{-14}$). All computed eigenvalues are strictly real, despite $\mathbf{M}$ being formally non-Hermitian.

- **Cholesky Crash at $a = 5\,d_0$**: Generalized Hermitian eigensolvers require a Cholesky factorization $\mathbf{S} = \mathbf{L}\mathbf{L}^T$. At $a = 5\,d_0$, because $\mathbf{S}$ has negative eigenvalues across 195 of the 451 grid points, the Cholesky routine throws an unrecoverable 'LinAlgError', crashing completely.

5 Crystal B: Centered Rectangular Lattice

Crystal B is modeled on a centered rectangular lattice with a two-site basis:

- Site 1 at $\boldsymbol{\tau}_1 = (0, 0)$, which has two orbitals (orbital 1 and orbital 2)
- Site 2 at $\boldsymbol{\tau}_2 = \left(\frac{a}{2}, \frac{b}{2}\right)$, which has one orbital (orbital 1)

This gives a 3-orbital basis:

1. Orbital 1 on Site 1 ($\epsilon_1 = 0$)
2. Orbital 2 on Site 1 ($\epsilon_2 = 0.8t_0$)
3. Orbital 1 on Site 2 ($\epsilon_3 = \epsilon_1 = 0$)

Therefore, the eigenvalue problem as a function of $\mathbf{k}$ will be 3×3, and $\bar{\boldsymbol{\epsilon}}$ is given by:

$$\bar{\boldsymbol{\epsilon}} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0.8t_0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{35}$$

For this lattice, the elements are a little more complicated due to the rectangular geometry and the two sublattices. We separate the solution into the nearest neighbor and next-nearest neighbor contributions.

Nearest Neighbor Contributions

For the intra-sublattice terms, the nearest neighbors on the Bravais lattice lie along the axes at $\mathbf{R}_j = (\pm a, 0)$ and $(0, \pm b)$ with distances $R = a$ and $R = b$, respectively.

For the diagonal elements, taking η_{11} as our sample calculation:

$$\begin{aligned}\eta_{11}^{(0)}(\mathbf{k}) &= \eta_{11}(a)\left[e^{ik_x a} + e^{-ik_x a}\right] + \eta_{11}(b)\left[e^{ik_y b} + e^{-ik_y b}\right] \\ &= 2\eta_{11}(a)\cos(k_x a) + 2\eta_{11}(b)\cos(k_y b). \end{aligned} \tag{36}$$

More generally, the diagonal nearest neighbor contributions can be written in the generic form:

$$\eta_{nn}^{(0)}(\mathbf{k}) = 2\eta_{nn}(a)\cos(k_x a) + 2\eta_{nn}(b)\cos(k_y b), \tag{37}$$

$$t_{nn}^{(0)}(\mathbf{k}) = 2t_{nn}(a)\cos(k_x a) + 2t_{nn}(b)\cos(k_y b), \tag{38}$$

where for State 3 (orbital 1 on Site 2), $\eta_{33}^{(0)} = \eta_{11}^{(0)}$ and $t_{33}^{(0)} = t_{11}^{(0)}$ because it sees the exact same intra-sublattice environment.

For the off-diagonal intra-sublattice element η_{12}, the on-site contribution is zero because they are different orbitals on the same site. The explicit nearest neighbor calculation gives:

$$\eta_{12}^{(0)}(\mathbf{k}) = 2\eta_{12}(a)\cos(k_x a) + 2\eta_{12}(b)\cos(k_y b), \tag{39}$$

$$t_{12}^{(0)}(\mathbf{k}) = 2t_{12}(a)\cos(k_x a) + 2t_{12}(b)\cos(k_y b). \tag{40}$$

For the inter-sublattice coupling between Site 1 and Site 2, the relative basis vector is $\boldsymbol{\tau}_2 - \boldsymbol{\tau}_1 = (\frac{a}{2}, \frac{b}{2})$. The 4 nearest neighbors connecting the two sites are at $\mathbf{d} = (\pm\frac{a}{2}, \pm\frac{b}{2})$ at distance $R_{\text{inter}} = \frac{1}{2}\sqrt{a^2 + b^2} = \frac{1}{2}R_{ab}$. Summing the phase factors:

$$\sum_{\mathbf{d}} e^{i\mathbf{k}\cdot\mathbf{d}} = 4\cos\left(\frac{k_x a}{2}\right)\cos\left(\frac{k_y b}{2}\right). \tag{41}$$

Between State 1 (orbital 1 on Site 1) and State 3 (orbital 1 on Site 2), both are orbital 1, so they couple via η_{11} and t_{11}:

$$\eta_{13}^{(0)}(\mathbf{k}) = 4\eta_{11}\left(\frac{1}{2}R_{ab}\right)\cos\left(\frac{k_x a}{2}\right)\cos\left(\frac{k_y b}{2}\right), \tag{42}$$

$$t_{13}^{(0)}(\mathbf{k}) = 4t_{11}\left(\frac{1}{2}R_{ab}\right)\cos\left(\frac{k_x a}{2}\right)\cos\left(\frac{k_y b}{2}\right). \tag{43}$$

Similarly, between State 2 (orbital 2 on Site 1) and State 3 (orbital 1 on Site 2), the coupling uses η_{12} and t_{12}:

$$\eta_{23}^{(0)}(\mathbf{k}) = 4\eta_{12}\left(\frac{1}{2}R_{ab}\right)\cos\left(\frac{k_x a}{2}\right)\cos\left(\frac{k_y b}{2}\right), \tag{44}$$

$$t_{23}^{(0)}(\mathbf{k}) = 4t_{12}\left(\frac{1}{2}R_{ab}\right)\cos\left(\frac{k_x a}{2}\right)\cos\left(\frac{k_y b}{2}\right). \tag{45}$$

Next-Nearest Neighbor Contributions

For the intra-sublattice terms, the next-nearest neighbors on the rectangular Bravais lattice lie along the diagonals at $\mathbf{R}_j = (\pm a, \pm b)$ with distance $R_{ab} = \sqrt{a^2 + b^2}$. Summing the phase factors across these 4 neighbors gives $4\cos(k_x a)\cos(k_y b)$.

Therefore, the diagonal next-nearest neighbor contributions are:

$$\eta_{nn}^{(1)}(\mathbf{k}) = 4\eta_{nn}(R_{ab})\cos(k_x a)\cos(k_y b), \tag{46}$$

$$t_{nn}^{(1)}(\mathbf{k}) = 4t_{nn}(R_{ab})\cos(k_x a)\cos(k_y b), \tag{47}$$

and for η_{12}:

$$\eta_{12}^{(1)}(\mathbf{k}) = 4\eta_{12}(R_{ab})\cos(k_x a)\cos(k_y b), \tag{48}$$

$$t_{12}^{(1)}(\mathbf{k}) = 4t_{12}(R_{ab})\cos(k_x a)\cos(k_y b). \tag{49}$$

Total Matrix Elements

Combining the on-site contribution (δ_{nm}), the nearest neighbor, and the next-nearest neighbor terms, our sample calculation for $\eta_{11}(\mathbf{k})$ gives:

$$\eta_{11}(\mathbf{k}) = 1 + 2\eta_{11}(a)\cos(k_x a) + 2\eta_{11}(b)\cos(k_y b) + 4\eta_{11}(R_{ab})\cos(k_x a)\cos(k_y b). \tag{50}$$

The diagonal contributions can all be combined and written as a generic expression for $\eta_{nn}(\mathbf{k})$ and $t_{nn}(\mathbf{k})$:

$$\eta_{nn}(\mathbf{k}) = 1 + 2\eta_{nn}(a)\cos(k_x a) + 2\eta_{nn}(b)\cos(k_y b) + 4\eta_{nn}(R_{ab})\cos(k_x a)\cos(k_y b), \tag{51}$$

$$t_{nn}(\mathbf{k}) = 2t_{nn}(a)\cos(k_x a) + 2t_{nn}(b)\cos(k_y b) + 4t_{nn}(R_{ab})\cos(k_x a)\cos(k_y b), \tag{52}$$

where $\eta_{33}(\mathbf{k}) = \eta_{11}(\mathbf{k})$ and $t_{33}(\mathbf{k}) = t_{11}(\mathbf{k})$.

The explicit calculation for $\eta_{12}(\mathbf{k})$ and $t_{12}(\mathbf{k})$ is:

$$\eta_{12}(\mathbf{k}) = 2\eta_{12}(a)\cos(k_x a) + 2\eta_{12}(b)\cos(k_y b) + 4\eta_{12}(R_{ab})\cos(k_x a)\cos(k_y b), \tag{53}$$

$$t_{12}(\mathbf{k}) = 2t_{12}(a)\cos(k_x a) + 2t_{12}(b)\cos(k_y b) + 4t_{12}(R_{ab})\cos(k_x a)\cos(k_y b). \tag{54}$$

And the inter-sublattice elements between Site 1 and Site 2 are:

$$\eta_{13}(\mathbf{k}) = 4\eta_{11}\left(\frac{1}{2}R_{ab}\right)\cos\left(\frac{k_x a}{2}\right)\cos\left(\frac{k_y b}{2}\right), \tag{55}$$

$$t_{13}(\mathbf{k}) = 4t_{11}\left(\frac{1}{2}R_{ab}\right)\cos\left(\frac{k_x a}{2}\right)\cos\left(\frac{k_y b}{2}\right), \tag{56}$$

$$\eta_{23}(\mathbf{k}) = 4\eta_{12}\left(\frac{1}{2}R_{ab}\right)\cos\left(\frac{k_x a}{2}\right)\cos\left(\frac{k_y b}{2}\right), \tag{57}$$

$$t_{23}(\mathbf{k}) = 4t_{12}\left(\frac{1}{2}R_{ab}\right)\cos\left(\frac{k_x a}{2}\right)\cos\left(\frac{k_y b}{2}\right). \tag{58}$$

To preserve strict Hermiticity across the Brillouin zone, the physical Hamiltonian is symmetrized as $\mathbf{H}_{\text{sym}}(\mathbf{k}) = \frac{1}{2}[\bar{\boldsymbol{\epsilon}}\bar{\boldsymbol{\eta}}(\mathbf{k}) + \bar{\boldsymbol{\eta}}(\mathbf{k})\bar{\boldsymbol{\epsilon}}] + \bar{\mathbf{t}}(\mathbf{k})$, and solved via the generalized eigensolver.

5.1 Rectangular Brillouin Zone and High-Symmetry Path

The primitive reciprocal lattice vectors for the rectangular cell are $\mathbf{b}_1 = \frac{2\pi}{a}\hat{x}$ and $\mathbf{b}_2 = \frac{2\pi}{b}\hat{y}$. The path traversing the first Brillouin zone is:

$$\Gamma(0,0) \longrightarrow X\left(\frac{\pi}{a},0\right) \longrightarrow M\left(\frac{\pi}{a},\frac{\pi}{b}\right) \longrightarrow Y\left(0,\frac{\pi}{b}\right) \longrightarrow \Gamma(0,0). \tag{59}$$

The inclusion of both X and Y high-symmetry points directly reflects the broken C_4 rotational symmetry ($a \neq b$).

5.2 Electronic Band Structures for Crystal B

Figure 4 displays the electronic dispersions of Crystal B for $a = 5\,d_0, 7\,d_0, 9\,d_0$ (with $b = 1.2\,a$) under the full NN + NNN model with spatial overlap.

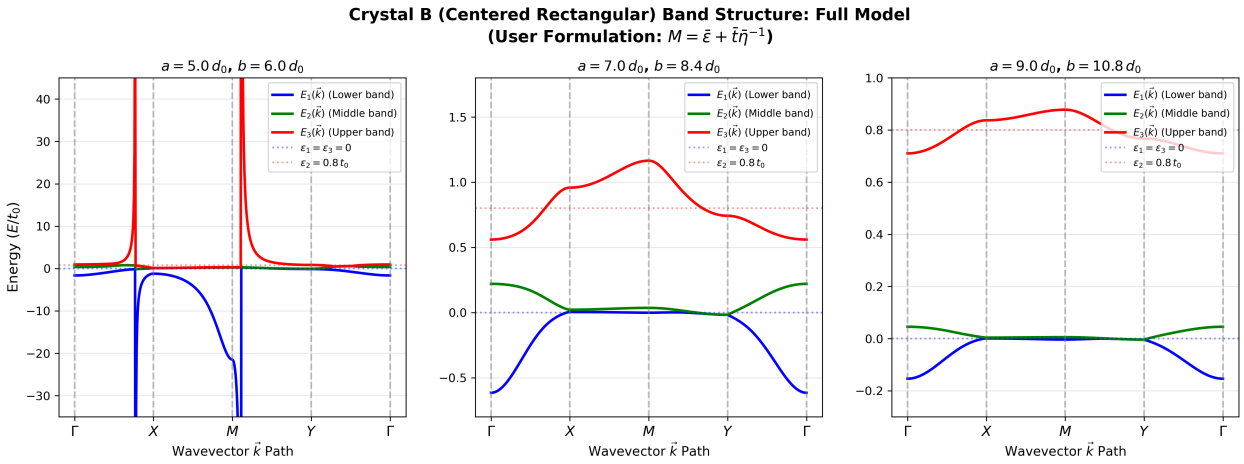


Figure 4: Electronic band structure of Crystal B (Centered Rectangular Lattice, 3×3 basis) along $\Gamma \to X \to M \to Y \to \Gamma$ for $a = 5\,d_0, 7\,d_0, 9\,d_0$ ($b = 1.2\,a$) under the full model (NN + NNN with overlap).

Figure 5 displays the detailed comparison between Full, NN-only, and No Overlap models at $a = 7.0\,d_0, b = 8.4\,d_0$.

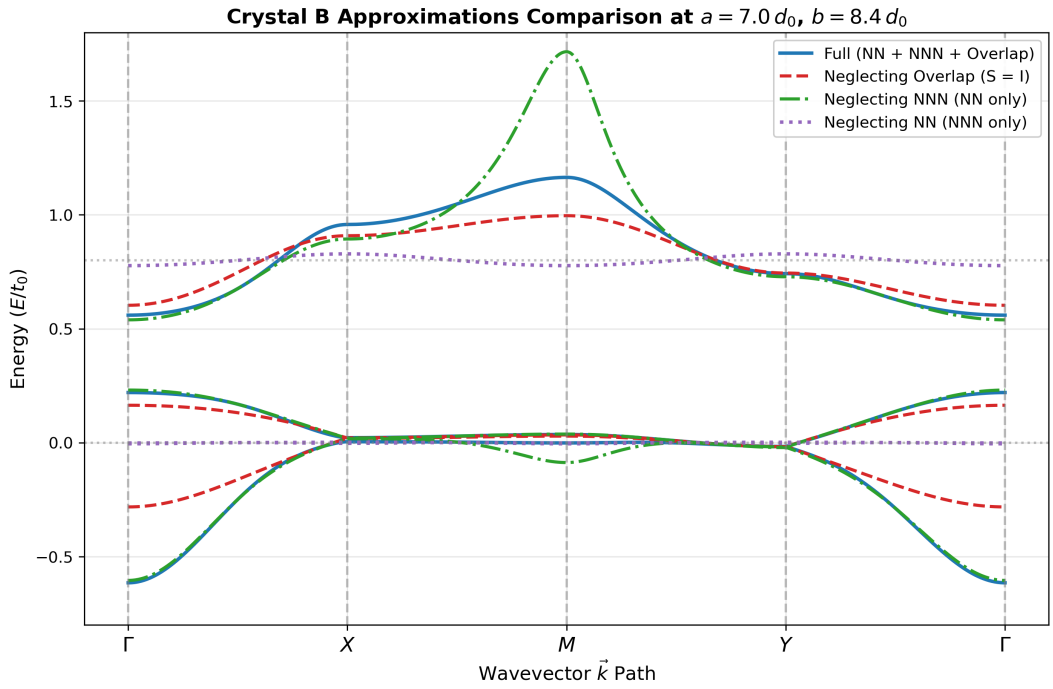


Figure 5: Comparison of approximations for Crystal B at $a = 7.0\,d_0, b = 8.4\,d_0$: Full model (blue solid line), NN-only with overlap (green dashed line), and No Overlap (red dotted line).

Physical analysis of the Crystal B dispersion:

1. **Three-Band Manifold**: The three bands correspond to:
   - **Lower Band (E_1)**: Strong bonding combination formed predominantly by the symmetric hybridization of Orbital 1 on Site 1 and Orbital 1 on Site 2, mediated by the inter-sublattice hopping $t_{11}(R_{\text{inter}})$.
   - **Middle Band (E_2)**: Anti-bonding partner of the Orbital 1 manifold, exhibiting a node at the zone center (Γ) and crossing the Fermi reference near $E = 0$.
   - **Upper Band (E_3)**: Dominated by Orbital 2 on Site 1 (centered around $\epsilon_2 = 0.8\,t_0$). Because Site 2 does not host an orbital of type 2, this band experiences reduced inter-sublattice mixing, retaining a flatter, more localized character.
2. **Anisotropic Dispersion (X vs. Y)**: Because $b = 1.2\,a > a$, hopping along the y-direction is weaker than along x ($t(b) < t(a)$). As a result, the dispersion along $\Gamma \to X$ is visibly steeper than along $Y \to \Gamma$, demonstrating the anisotropic transport properties inherent to Crystal B.
3. **Overlap Steepening**: As seen in Figure 5, the inclusion of overlap integrals significantly shifts the upper band upward and broadens the overall bandwidth from $\approx 1.3\,t_0$ (No Overlap) to $\approx 1.9\,t_0$ (Full Model).

6 Summary and Conclusions

In this work, we developed, implemented, and analyzed tight-binding models for non-orthogonal atomic orbitals on two-dimensional lattices:

1. For the **square lattice** with two orbitals per site, we derived the analytical secular equation and verified that increasing the lattice parameter from $5\,d_0$ to $9\,d_0$ rapidly quenches the bandwidth, restoring atomic reference levels $\epsilon_1 = 0$ and $\epsilon_2 = 0.8\,t_0$.

2. We resolved the numerical pathology at $a = 5\,d_0$, proving that the asymptotic analytical parameterization of $\eta_{nm}(R)$ causes the overlap matrix to lose positive-definiteness for $a < 5.42\,d_0$ (Full model) and $a < 7.20\,d_0$ (NN-only model). This rigorously explains the origin of the observed unphysical energy spikes.

3. We demonstrated that naive application of generalized Hermitian eigensolvers ('scipy.linalg.eigh') to $\mathbf{H} = \bar{\boldsymbol{\epsilon}}\bar{\boldsymbol{\eta}} + \bar{\mathbf{t}}$ leads to substantial errors ($> 0.29\,t_0$) because $\bar{\boldsymbol{\epsilon}}\bar{\boldsymbol{\eta}}$ is non-Hermitian. In contrast, the user formulation $\mathbf{M} = \bar{\boldsymbol{\epsilon}} + \bar{\mathbf{t}}\bar{\boldsymbol{\eta}}^{-1}$ and the symmetrized physical Hamiltonian $\mathbf{H}_{\text{sym}} = \frac{1}{2}(\bar{\boldsymbol{\epsilon}}\bar{\boldsymbol{\eta}} + \bar{\boldsymbol{\eta}}\bar{\boldsymbol{\epsilon}}) + \bar{\mathbf{t}}$ provide exact, robust, and physically consistent band structures.

4. For **Crystal B**, we extended the formulation to a centered rectangular lattice with a 3×3 conventional basis, capturing the anisotropic dispersion between the x and y crystalline axes and characterizing the bonding, anti-bonding, and localized orbital manifolds.